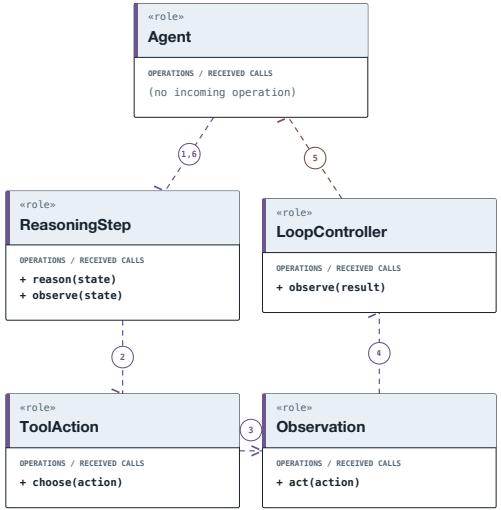
R01 Reason-act-observe

Reasoning / UML CLASS VIEW / ANALYTICAL MODEL

Interleave reasoning and external actions.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	An agent can reason and use an external action interface.
TRIGGER	The model emits reasoning followed by an action request.
STRUCTURE	Parsed action, execution result, and observation fed into the next cycle.
EFFECT	External evidence enters the reasoning trajectory before the next action.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Action and observation schemas are stable enough to parse.
ENSURES	Every executed action has an attributable observation or execution error.
INVARIANT	Model reasoning is not treated as environment evidence.
VIOLATION	Invalid action, unavailable tool, or parse error is exposed to the loop.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 Agent → ReasoningStep reason(state)
2 ReasoningStep → ToolAction choose(action)
3 ToolAction → Observation act(action)
4 Observation → LoopController observe(result)
5 LoopController ↩ Agent continue(observation)
6 Agent → ReasoningStep observe(state)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Prompt protocol, typed tool call, XML or text parser, and observation compression.
TEST_OBLIGATION	Test action-observation linkage, malformed action, tool error, and loop termination.
NEGATIVE_CONTROL	Chain-of-thought text without external action is not ReAct.